

PHYSICS FOR ENGINEERS I

PHY1163

Department of Physics

July 8, 2022

QUIZ /10

Q1 Velocity and acceleration are both vectors.

1. Are the velocity and the acceleration always in the same direction? [1]
2. Is it possible for an object to have a zero velocity and a non-zero acceleration? [1]

Q2 The ideal gas law is given by the relation

$$PV = nRT, \quad (1)$$

where, P, V are respectively the pressure and volume of the gas, n the quantity of matter, R the ideal gas constant and T the absolute temperature.

1. From (1), find the dimensions and SI units of R , [2]
2. Is R a dimensionless constant? [1]
3. The equation (1) can be expressed as $PV = Nk_B T$, where N is the number of molecules of gas and k_B is the Boltzmann constant. Find the dimensions and SI units of k_B . [2]

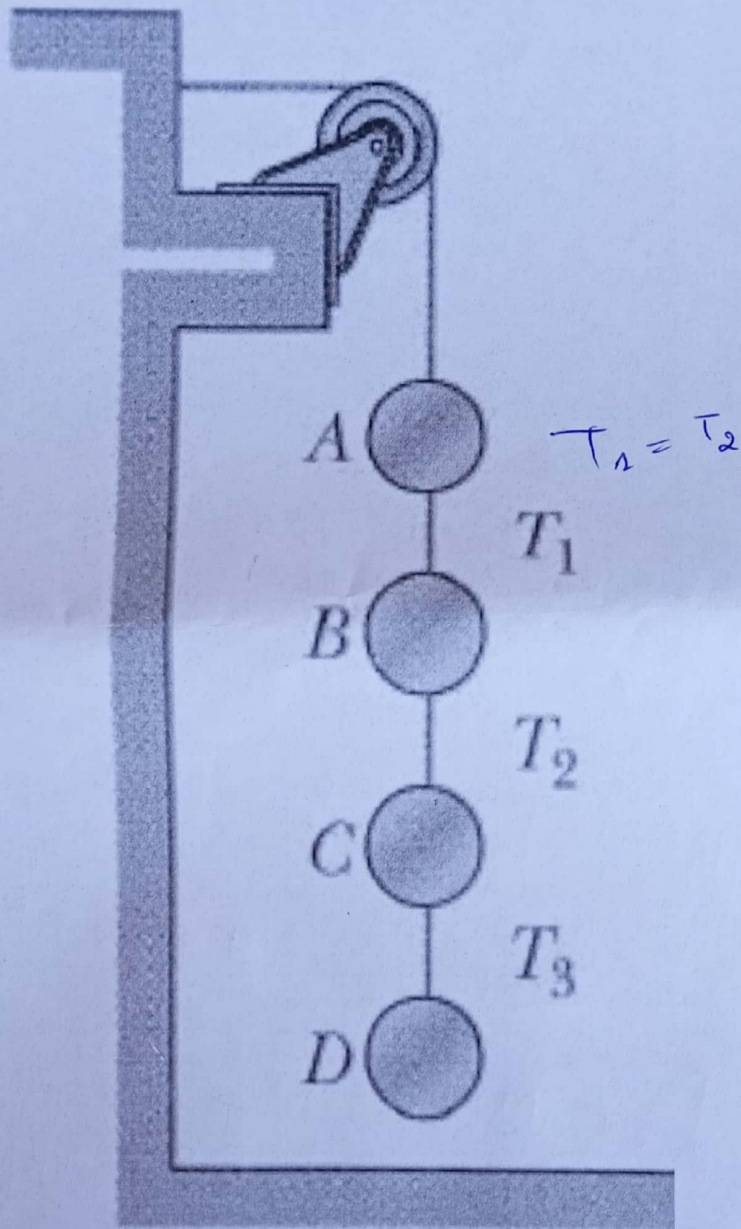
Q3 Figure 3 shows an arrangement in which four disks are suspended by cords. The longer, top cord loops over a frictionless pulley and pulls with a force of magnitude 98 N on the wall to which it is attached. The tensions in the shorter cords are $T_1 = 58.8$ N, $T_2 = 49.0$ N, and $T_3 = 9.8$ N. What are the masses of (a) disk A, (b) disk B, (c) disk C, and (d) disk D? [3 Marks]

$$\begin{array}{r} 81 \\ 67.5 \\ 42.8 \\ \hline 267 \end{array}$$

$$\begin{array}{r} 71 \\ 83.8 \\ - 69.5 \\ \hline 14.3 \end{array}$$

$$F = ma..$$

$$T = FR$$



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DEPARTMENT OF PHYSICS

CONTINUOUS ASSESSMENT TEST 1

ACADEMIC YEAR: 2021-2022

STUDY GROUP: Y1 MEE(All),SGE, Geotechnical

MODULE NAME: PHYSICS FOR ENGINEERS I(PHY1163)

DATE: July 13, 2022

MAXIMUM MARKS: 20

DURATION: 1.5 hours

QUESTION N^o1 [12 Mark]

a) Read the statements carefully and answer **TRUE** if it is correct or **FALSE** otherwise and Explain your reasoning.

Velocity and acceleration are both vectors.

1^o. The velocity and the acceleration are always in the same direction.

2^o. It is possible for an object to have a zero velocity and a non-zero acceleration.

b) Abia walk 73.2 m at a speed of 1.22m/s and then run 13.2m at a speed of 3.05 m/s along a straight track. (1^o.) Compute your average velocity for Abia's trajectory. (2^o.) Graph x versus t for this trajectory and indicate how the average velocity is found on the graph.

c) A particle moves in the xy plane (Figure 1), starting from the origin at $t = 0$ with an initial velocity having an x component of 20 m/s and a y component of -15 m/s. The particle experiences an acceleration in the x direction, given by $a_x = 4.0\text{m/s}^2$.

1^o. Determine the total velocity vector at any time.

2^o. Calculate the velocity and speed of the particle at $t = 5.0$ s and the angle the velocity vector makes with the x axis.

3^o. Determine the x and y coordinates of the particle at any time t and its position vector at this time.

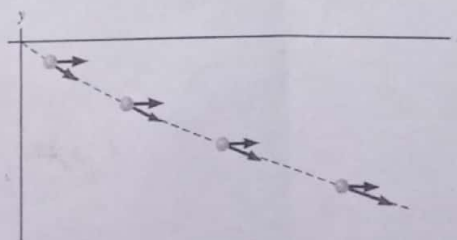


Figure 1: Motion diagram for the particle.

QUESTION N^o2 [8 Mark]

a) Define:

- Torque
- Rigid body
- Angular momentum
- Free body diagram

b) Consider two equal masses (3 kg) at the end of a bar (with negligible mass) of 4 m long rotating about a specified axis.
Calculate:

1. The moment of inertia when they are rotating around the center of their rod.
2. The moment of inertia when they are rotating at the end of the rod.
3. The moment of inertia when they are rotating at a point 2 m from one end of the rod.
4. Rework (2) and (3) using Steiner's theorem.

GOOD LUCK!!!!

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CONTINUOUS ASSESSMENT TEST-CAT

MODULE: PHY1163-Physics for Engineers I /40marks

TARGET GROUPS: EPE, Level 1

Date: 13th July, 2022

Duration: 2hours

QUESTION N^o 1 /15marks

- a) In the freely falling motion a ball is thrown vertically upward. What are its velocity and acceleration when it reaches its maximum altitude? What is its acceleration just before it strikes the ground? /3marks
- b) A stone is thrown from the top of a building with an initial velocity of 20m.s^{-1} straight upward. The building is 50 m high, and the stone just misses the edge of the roof on its way down. Determine: /4marks
- The time needed for the stone to reach its maximum height,
 - The maximum height,
 - The velocity and position of the stone at $t = 5\text{s}$
- c) A stone is catapulted at time $t = 0\text{s}$, with an initial velocity of magnitude 20.0 m/s and at an angle of 40.0° above the horizontal.
- What are the horizontal x - and vertical y - components of its displacement from the catapult site (x_0, y_0) at time $t = 1.10\text{s}$? /4marks
 - How far will the stone travel in the horizontal direction at that time? /2marks
 - How long will it take the stone to turn back to the level of catapult site? /2marks

QUESTION N^o 2 /10marks

- A) Starting with the formula for the moment of inertia of a long thin rod rotated around an axis passing at one end perpendicular to its length $I = \frac{1}{3}ML^2$, prove that the moment of inertia of the same long thin rod rotated about an axis passing through its center perpendicular to its length is $I = \frac{1}{12}ML^2$. Where M is the mass of the rod, and L is its the length. /5marks
- B) A spherical ball of mass $M = 7.2\text{kg}$ and radius $R = 11\text{cm}$, starts from rest state and rolls without friction on a inclined plane by $\theta = 34^\circ$ to the horizontal and of length $l = 2.1\text{m}$. Determine:

Diagram of a rod of length L with an axis of rotation at one end.

$$A = L \times W$$

$$y = 20 \times 5 - \frac{1}{2} \times 9.81 \times 25$$

$$y = 100 - 122.625$$

$$A = \pi r^2 l$$

$$dm = \frac{2}{r} M$$

$$\Rightarrow \frac{dm}{M} = \frac{2}{r}$$

- a) The linear velocity V_C of the ball when it reaches the bottom of the slope. (**Hint:** use concept of the conservation of total mechanical energy, and noting that $I_C = \frac{2}{5}MR^2$) /3marks
- b) The total kinetic energy K of the ball at that position. /2marks

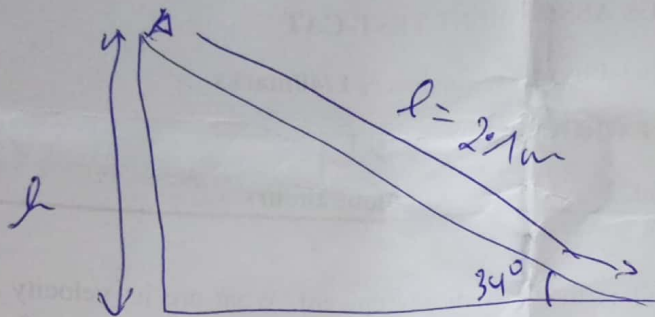


Figure 1

$$\omega = \frac{v}{r}$$

$$M = PE - x k \cdot v$$

QUESTION N° 3 / 15marks

- A) State the work-energy theorem and give its mathematical expression. /2 marks
- B) Using the relationship between the conservative force and the potential energy function, find the conservative forces for the following potential energy functions: /3marks
- $U(x) = \frac{1}{2} kx^2$
 - $U(y) = mgy$
- C) A man pulls a wooden box on a frictionless floor with a force F of magnitude 50 N at an angle of 30° with the horizontal as shown in figure below.
- Draw a labeled free body diagram showing all the forces that act on the wooden box. /2marks
 - What is the magnitude of the work done by the weight of the wooden box? Explain briefly /3marks
 - Calculate the work done by the man on the wooden box as it is displaced 3 m to the right /5marks

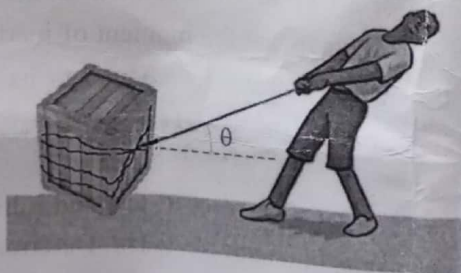


Figure 2

$$W = \int F dx$$

$$W =$$

$$-F_x x = \frac{1}{2} kx^2$$

$$-F_x dx = \frac{1}{2} k dx^2$$

$$-F_x = kx$$

$$F_x = -kx$$

$$W = \frac{f}{x}$$

END OF QUESTIONS



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PHYSICS FOR ENGINEERS I (PHY1163)

Programme: Y1 ETE

Date: 15/07/2022

Duration : 2hours

Max: 30

Question 1 (10 Marks)

(a) Determine whether the dimensions in each of the following equations are consistent. With the variable parameters v , a , t , x , m , and F represent respectively, velocity, acceleration, time, distance, mass, and Force.

(i) $v^2 = at$

(ii) $x = vt + \frac{1}{2}at^2$

(iii) $max = vt + \frac{1}{2}at^2$

(iv) $v^2 = 2ax + \frac{x}{t}$

(v) ~~Fx~~ $= 2ax + \frac{x}{t}$

(b) The position of a particle moving along the x-axis depends on the time according to the equation $x = ct^2 - bt^3$, where x is in meters and t in seconds.

1. What are the units of:

(i) constant c and (ii) constant b

2. Let their numerical values be 3.0 and 2.0, respectively. What is the particle's average velocity between $t=1.0$ s and $t= 3.5$ s?

3. At what time does the particle reach its maximum positive x position?

4. What is the particle's acceleration at $t = 2.50$ s?

(c) The acceleration of a 1-dimensional particle is proportional to its speed, $a = -kv$, with $k > 0$ is constant. Find the position of the particle as function of time t : $x = x(t)$, given a nonzero initial speed, $v_0 = v(t = 0)$

Question 2 (8 Marks)

State the Newton's laws. A block of mass $m = 2.00$ kg is released from rest at $h = 0.500$ m above the surface of a table, at the top of a $\theta = 30.0$ incline as shown in Figure below. The frictionless incline is fixed on a table of height $H = 2.00$ m.

(1) Determine the acceleration of the block as it slides down the incline.

- (2) What is the velocity of the block as it leaves the incline?
- (3) How far from the table will the block hit the floor?
- (4) How much time has elapsed between when the block is released and when it hits the floor?
- ✓ (5) Does the mass of the block affect any of the above calculations? Refer to the figure 1 below:

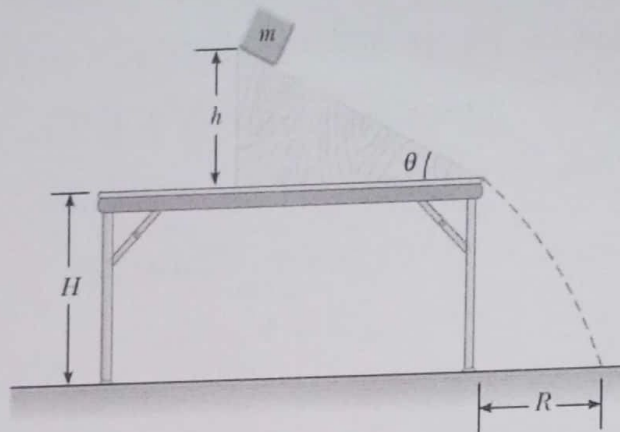


Figure 1: Fig. 1

Question 3 (12 Marks)

1. A 12kg mass under the influence of a force $\vec{F} = (4t^2\vec{i} - 4t\vec{j})$ N, where t is time in seconds. It starts at rest from the origin at $t = 0$. Find: (i) Its velocity, (ii) Its position, (iii) the torque, and (iv) Calculate $\vec{r} \times m\vec{v}$ where m is the mass. What physical quantity is that?
2. State the conditions under which a rigid body is in equilibrium. Where must a 0.80-kN object be hung on a uniform, horizontal, rigid 100-N pole so that a girl pushing at one end supports one-third as much as a woman pushing up at the other end?
3. (i) State the four (4) assumptions based on when deriving the continuity equation along a streamline.
- (ii) In Figure 2, water flows through a horizontal pipe and then out into the atmosphere at a speed $v_1 = 15\text{ m/s}$. The diameters of the left and right sections of the pipe are 5.0 cm and 3.0 cm. (a) What volume of water flows into the atmosphere during a 10 min period? and the speed v_2 in the left section.

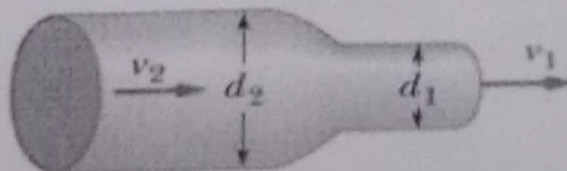


Figure 2: Fig. 2

Best of Luck

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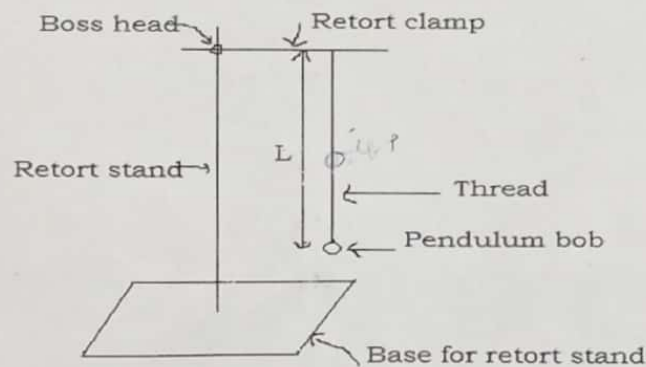
School of Science

Department of Physics

EXPERIMENT N0-1: Determination of the acceleration due to gravity

PROCEDURES

- a) Suspend the pendulum bob by means of a thread from a retort stand as shown below



- b) Adjust the length L of the pendulum to 0.45 m
- c) Displace the pendulum bob through a small angle and let it oscillate freely.
Time 20 complete oscillations and record the time t
- d) Repeat the procedures (b) to (c) for length L of the simple pendulum equals to 0.50, 0.55, 0.60, 0.65, 0.70 m
- e) Tabulate your results in the following table including the time t for 20 oscillations, the time T for one complete oscillation and T^2 .

All physical quantities must have two decimal places

L/m	t/s	T/s	T ² /s ²	g/ms ⁻²
0.45				
0.50				
0.55				
0.60				
0.65				
0.70				

Remember that $T = 2\pi \sqrt{\frac{l}{g}}$

I First method

- Compute the average acceleration due to gravity using the formula $\bar{g} = \frac{1}{N} \sum g_i$
- Find the relative uncertainty of the acceleration $\frac{\Delta g}{g}$ by the formula

$$\frac{\Delta g}{g} = \sqrt{\left(\frac{\Delta L}{L}\right)^2 + \left(\frac{2\Delta T}{T}\right)^2}$$
, where $\Delta L = 0.5\text{cm}$ and $\Delta T = 0.2\text{s}$
- Compute the standard deviation in both length and T²
- Write down the Linear curve fitting (Least square fitting)'s equation.
- Write down the experimental result of the acceleration due to gravity.

II Second method

- Plot a graph of length against the square of period
- Determine the gradient from your graph
- Determine the value of the acceleration due to gravity using $g = 4\pi^2 s$, where s is the gradient obtained from the graph
- Compare your findings [one from (h), and other from (k)] with the accepted value $g = 9.8\text{m/s}^2$
- Repeat procedure (c) to (e) using pendulum bob of mass different to what you used before.
- Make your conclusion.